You should therefore prefer using the variants with a 128-bit authentication tag unless you have other measures in place to prevent message forgery, such as strong network protections, and you know that you need to reduce network overheads. You should apply strict rate-limiting to API endpoints where there is a risk of brute force attacks against authentication tags; see chapter 3 for details on how to apply rate-limiting.

12.2 Pre-shared keys

In some particularly constrained environments, devices may not be capable of carrying out the public key cryptography required for a TLS handshake. For example, tight constraints on available memory and code size may make it hard to support public key signature or key-agreement algorithms. In these environments, you can still use TLS (or DTLS) by making use of cipher suites based on *pre-shared keys* (PSK) instead of certificates for authentication. PSK cipher suites can result in a dramatic reduction in the amount of code needed to implement TLS, as shown in figure 12.5 because the certificate parsing and validation code, along with the signatures and public key exchange modes can all be eliminated.